Big Data Course

Capstone Project
Work Breakdown Structure

For students

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Capstone Project Work Breakdown Structure (WBS)

Today	**10/14/2026**
Project Team Name	**Team 4**
Project Topic	**SentinelGuard: Real-time Sentiment Classification of Live Chat Comments for Early Detection of Harmful Comment Spikes**

Task		Start date	End date	Responsibility	Deliverable	Task Status	W1					W2					W3					W4				
							D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	D12	D13	D14	D15	D16	D17	D18	D19	D20
							9/15	9/16	9/17	9/18	9/19	9/20	9/21	9/22	9/23	9/24	9/25	9/26	9/27	9/28	9/29	10/30	10/1	10/2	10/3	10/4
1. Project planning and management						In-progress																				
	Team setup and topic framing	9/15/2026	9/16/2026	Ky	docs/topic_decision.md	Done																				
	Action Plan preparation	9/16/2026	9/17/2026	Ky, Dâm	deliverables/Action_Plan.docx	Done																				
	Data selection and source/licence verification	9/16/2026	9/18/2026	Hạnh	data/raw/SOURCE.md	Done																				
	Work Breakdown Structure (WBS) preparation	9/18/2026	9/18/2026	Ky	deliverables/WBS.xlsx	In-progress																				
	Preliminary presentation (Day 5)	9/18/2026	9/19/2026	All members	Proposal slides	Delayed																				
	WBS revision for review checkpoints	9/23/2026	9/24/2026	Ky	Updated WBS.xlsx (checks Day 9, 14, 19)	Delayed																				
2. Data preparation						Delayed																				
	Data quality assessment (duplicates, rating distribution)	9/20/2026	9/21/2026	Hạnh	Quality report table	Delayed																				
	Shared text-cleaning module with unit tests	9/20/2026	9/21/2026	Dâm	src/common/text_cleaning.py	Delayed																				
	Rating-to-sentiment mapping, dedup, fixed split	9/21/2026	9/22/2026	Hạnh	data/processed/ + eval_labels.parquet	Delayed																				
3. Data ingestion pipeline (Kafka)						Delayed																				
	Kafka setup, topics and configuration	9/22/2026	9/23/2026	Bằng	kafka/ scripts + README	Delayed																				
	Replay producer (rate, burst, late records)	9/23/2026	9/24/2026	Bằng	src/producer/replay_producer.py	Delayed																				
	Ingestion throughput measurement	9/24/2026	9/24/2026	Bằng	outputs/tables/ingestion_*.csv	Delayed																				
4. Data transformation (Spark Structured Streaming)						Delayed																				
	Streaming job: explicit schema, cleaning, DLQ	2026-09-24	2026-09-25	Dâm	src/streaming/stream_job.py	Delayed																				
	Event-time windows and watermark aggregation	2026-09-25	2026-09-26	Dâm	Parquet aggregates	Delayed																				
	Fault-tolerance test (restart mid-stream)	2026-09-26	2026-09-27	Dâm	Test results table	Delayed																				
5. Model training and evaluation						Delayed																				
	Spark ML pipeline, TF-IDF fitted on train only	2026-09-25	2026-09-26	Hạnh	src/training/train_model.py	Delayed																				
	Training and tuning with class weights	2026-09-26	2026-09-27	Hạnh	outputs/models/	Delayed																				
	Final test evaluation: macro-F1, per-class metrics	2026-09-27	2026-09-28	Hạnh	outputs/tables/metrics.csv	Delayed																				
6. Query, visualization and dashboard						Delayed																				
	Spark SQL queries and aggregation tables	2026-09-28	2026-09-29	Nguyên	src/eda/	Delayed																				
	Figures and insight notes	2026-09-29	2026-09-30	Nguyên	outputs/figures/, insights.md	Delayed																				
	Streamlit dashboard and alert threshold	2026-09-30	2026-10-01	Nguyên	src/app/app.py	Delayed																				
7. Prototype demonstration and delivery						Delayed																				
	Prototype showing presentation (Day 11)	2026-09-25	2026-09-25	All members	Prototype slides	Delayed																				
	Usage scenario and operating guide	2026-10-01	2026-10-02	Dâm, Ky	docs/scenario.md	Delayed																				
	Final report (Day 18)	2026-10-01	2026-10-02	All members	deliverables/Final_Report.docx	Delayed																				
	Final presentation and live demo (Day 20)	2026-10-05	2026-10-04	All members	Final slides	Delayed																				